Literature Review on Plant Disease Prediction Using Machine Learning and Deep Learning

This literature review explores recent advancements in plant disease prediction using machine learning and deep learning techniques. Plant diseases significantly impact agricultural productivity, and early detection is crucial for reducing crop losses. With the advancement of artificial intelligence, researchers have developed automated systems that analyze plant images and environmental data to detect and predict diseases with high accuracy.

Recent studies highlight the effectiveness of machine learning and deep learning approaches in crop disease diagnosis. A comprehensive review published in 2024 examines algorithms such as Support Vector Machines (SVM), Random Forest, K-Nearest Neighbors (KNN), and deep learning architectures like Convolutional Neural Networks (CNN), VGG16, and ResNet. The study concludes that deep learning models outperform traditional techniques in terms of accuracy and robustness.

A systematic review analyzing over 100 research articles demonstrates the increasing adoption of artificial intelligence in plant disease detection. The findings indicate that CNN-based models achieve the highest performance, especially in image-based classification tasks. Early detection using these models plays a key role in improving crop yield and preventing widespread infection.

Another research study compares multiple algorithms, including Random Forest, KNN, SVM, and CNN, for plant leaf disease recognition. The results show that CNN achieves the highest accuracy, exceeding 97%, due to its ability to learn complex visual features directly from images. In contrast, traditional methods require manual feature extraction and achieve comparatively lower performance.

A review published in 2022 emphasizes the importance of artificial intelligence in modern agriculture. It highlights that deep learning techniques provide more reliable and scalable solutions compared to classical machine learning methods. The use of image processing combined with AI enables precise identification of diseases at early stages.

A hybrid model combining CNN with Random Forest has also been proposed to enhance prediction performance. In this approach, CNN is used for feature extraction, while Random Forest performs classification. The hybrid model demonstrates improved accuracy compared to individual models, suggesting that combining techniques can yield better results.

Comparative studies on traditional algorithms such as Decision Tree, Naive Bayes, Random Forest, and SVM show that Random Forest often achieves higher accuracy in structured datasets. However, its performance is still lower than deep learning approaches when dealing with image data.

Another recent review outlines the role of classical algorithms like SVM, KNN, and Decision Trees in plant disease classification. These methods rely on features such as color, texture, and shape extracted from images. While effective, they are increasingly being complemented by deep learning models.

Key Algorithms Identified:

1. Convolutional Neural Networks (CNN) – highly effective for image-based disease detection.
2. Support Vector Machine (SVM) – widely used for classification tasks with extracted features.
3. Random Forest – ensemble method providing stable and reliable predictions.

4. K-Nearest Neighbors (KNN) – simple algorithm based on distance metrics.

Conclusion: The literature indicates that artificial intelligence has significantly improved plant disease prediction. While traditional machine learning algorithms remain useful, deep learning approaches, particularly CNN, provide superior performance. Hybrid models and advanced architectures are emerging as promising solutions for achieving higher accuracy and scalability in agricultural applications.